Khaled Hasan

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Website | **Google Scholar** | **GitHub**

Summary

Computer Science lecturer and researcher working on reliable foundation-model and agentic AI systems for real-world use. Current work spans multimodal multi-agent reasoning, LLM/VLM trust calibration, multilingual model reliability, and deployment-shift robustness, with emphasis on evaluation, uncertainty, and failure analysis.

Research Interests: LLM/VLM Systems, Agentic & Multi-Agent AI, Trustworthy AI, Multimodal AI, Uncertainty & Calibration

Education

Military Institute of Science and Technology (MIST) Jan. 2020 – Mar. 2024

B.Sc. in Computer Science and Engineering | CGPA: **3.83/4.00** | Rank: **4th of 88** Dhaka, Bangladesh

- **Thesis:** Adaptive User Interface in Mobile Banking: A Machine Learning Approach to Enhance UX [J1].
- **Advisors:** Asst. Prof. Iyolita Islam and Dr. Muhammad Nazrul Islam | Highest semester GPA: **3.98/4.00**.
- **English Proficiency:** IELTS Academic Overall **7.5**

Ongoing Research

- **Multimodal Agentic AI Systems**
  Developing a LangGraph-based multi-agent framework in which specialized agents process multimodal evidence, exchange context, coordinate reasoning, and perform cross-agent verification. The work studies information sharing, unreliable intermediate outputs, error propagation, and verification across agent workflows.
- **Human-Centered Trust Calibration in LLMs and VLMs**
  Investigating how model confidence, uncertainty, explanations, and high-confidence errors influence appropriate human reliance on language and multimodal AI systems.
- **Trustworthy Multilingual Prompting and Tokenization in LLMs**
  Conducting a cross-model study of Bangla, Banglish, and English prompting to examine how linguistic representation and tokenization affect response quality, hallucination, confidence, and reliability.
- **Reliable Edge AI under Distribution Shift**
  Investigating dependable machine learning as sensing conditions, deployment environments, and input distributions change, with emphasis on robust evaluation, uncertainty-aware adaptation, and reliable inference.
- **Trustworthy and Explainable Skin-Lesion Classification**
  Developing a DINO-ViT-based framework on HAM10000 incorporating uncertainty quantification, visual explanations, fairness analysis, and clinician-centered evaluation for reliable medical-image classification.

Research Experience

- **Research Collaborator, Reliable AI and Intelligent Systems** Nov. 2025 – Present
  Military Institute of Science and Technology (MIST) | Dhaka, Bangladesh
  Lead and co-supervise research from problem formulation through implementation, experimental evaluation, error analysis, interpretation, and manuscript preparation across AI systems, trustworthy machine learning, LLM/VLM systems, multimodal AI, and intelligent decision support.
  Current work includes multimodal agentic AI, trust calibration, multilingual LLM reliability, reliable learning under distribution shift, and explainable AI.
  Domains: AI Systems, LLM/VLM Systems, Agentic AI, Trustworthy Machine Learning, Multimodal AI.
- **Co-Investigator, Neuroadaptive AI from 32-Channel EEG** Mar. 2026 – Present
  MIST Research Grant | PI: Dr. T. M. Shahriar Sazzad | Dhaka, Bangladesh
  Developing a self-supervised Transformer pipeline over 32-channel EEG for modeling cognitive workload, vigilance, and affective states, with an interpretable closed-loop feedback architecture.
  Domains: Transformers, Multimodal Learning, Intelligent Sensing, Explainable AI.

- **Undergraduate Researcher** Jan. 2022 – Mar. 2024
  Military Institute of Science and Technology (MIST) | Dhaka, Bangladesh
  Conducted research spanning machine learning, semantic representation, adaptive intelligent interfaces, computer vision, IoT, and data-driven applications. Developed the adaptive mobile-banking interface published as [J1].
  Domains: Machine Learning, NLP, Human-Centered AI, Computer Vision, IoT.

Professional Experience

- **Lecturer, Department of Computer Science and Engineering** Apr. 2024 – Present | **Profile**
  Military Institute of Science and Technology (MIST) | Dhaka, Bangladesh
  Teach undergraduate courses in programming, computer architecture, operating systems, and networking; co-supervise research and thesis projects involving machine learning, trustworthy AI, intelligent systems, and applied computing.

Technical Skills

AI & Machine Learning: Machine Learning, Transformers, LLM/VLM Systems, Multimodal Learning, Agentic AI, Representation Learning

Reliability & Evaluation: Uncertainty and calibration analysis, hallucination/failure analysis, distribution-shift evaluation, cross-agent verification

Tools & Frameworks: Python, LangGraph, Sentence-BERT, OpenCV, SHAP

Publications

Peer-Reviewed Journal Article

[J1] **K. Hasan**, M. R. U. Islam, S. E. Arafat, and I. Islam. 2026. **Adaptive User Interface for Mobile Banking Apps: Enhancing UX through Machine Learning**. *Array* (Elsevier), 30, Article 100901. **[Q1, IF 4.5] DOI: 10.1016/j.array.2026.100901**

Peer-Reviewed Conference Papers

[C1] **K. Hasan**, A. Hossain, M. R. U. Islam, A. P. Saha, and F. Tazwar. 2025. **From Keywords to Context: Semantic (SBERT) vs. Traditional Matching for Job Recommendation**. IEEE ICCIT 2025. **DOI: 10.1109/ICCIT68739.2025.11491820**

[C2] M. R. U. Islam, **K. Hasan**, T. H. Rizu, I. Islam, and M. N. Islam. 2025. **Assessing Usability of Agricultural Mobile Applications for Illiterate and Semi-Literate Farmers in Bangladesh**. IEEE ICCIT 2025. **Top 2% Paper Award**. **DOI: 10.1109/ICCIT68739.2025.11491675**

[C3] **K. Hasan**, A. Kabir, H. Chowdhury, et al. 2025. **Evaluating User Experience in Decentralized Social Media Platforms: Comparative Study of Web3 and Web2 Social Media Apps**. HCI-E 2025; presented, to appear in Springer CCIS. **DOI: Not yet assigned**

[C4] **K. Hasan**, M. Zerin, E. A. Khan, et al. 2024. **Light Canvas: An IoT-Based Interactive Virtual Drawing System for Enhancing Educational Engagement**. ACM ICCA 2024. **DOI: 10.1145/3723178.3723192**

Manuscript Under Review

[R1] M. R. Mawla, R. Ryean, A. Rahman, et al., including **K. Hasan**. 2025. **Protective Zoning and Predictive Modeling for Nuclear Safety: A Machine Learning Approach at Rooppur**. Under review at the **Q1** *Journal of Nuclear Materials.* **(under review)**

Notable Projects

- **AutoML Tool for Air-Pollution and Disease-Burden Analysis** **GitHub**
  Developed a machine-learning framework using 34,920 observations from 194 countries. Country-grouped validation exposed generalization weaknesses hidden by random splitting, motivating a redesigned evaluation and SHAP-based interpretation.
  Focus: Python, Machine Learning, SHAP, Robust Evaluation, Distribution-Shift Analysis.

- **Semantic Job Recommendation Using SBERT** **GitHub** | **Paper**
  Developed a semantic recommendation framework using Sentence-BERT representations and compared contextual semantic modeling with conventional machine-learning approaches.
  Technologies: Python, Sentence-BERT, NLP, Representation Learning, Machine Learning.

- **Adaptive Mobile-Banking Interface Using Machine Learning** **Paper**
  Developed and evaluated a machine-learning-driven adaptive interface using user behavior for personalized adaptation, combining predictive modeling, clustering, SHAP-based explanation, and human-centered evaluation; published in Q1 *Array*.
  Focus: Human-Centered ML, Adaptive Interfaces, Explainable AI, SHAP, Statistical Evaluation.
- **Light Canvas: IoT-Based Interactive Virtual Drawing System** **GitHub** | **Video** | **Paper**
  Developed an end-to-end computer-vision and IoT system integrating visual tracking, sensing, image processing, application software, and real-time interaction; published at ACM ICCA 2024.
  Focus: Python, OpenCV, Computer Vision, IoT, Image Processing, Flutter.

Honors & Awards

- **Top 2% Paper Award** IEEE ICCIT 2025 Dec. 2025 | **Certificate**
- **Commandant's List of Honor** MIST; CGPA 3.80+ in 2020, 2022, and 2023 2020 – 2023 | **Official Link**
- **Dean's List of Honor** MIST; CGPA 3.70+ in all four academic years 2020 – 2024 | **Certificate**
- **Selected Project Demo (Team Lead)** BEAR Summit, National Semiconductor Symposium Jul. 2025 | **Link**
- **Champion, Research-Based Poster Presentation** BUP IEEE Branch Fest; Team Lead Aug. 2023 | **Certificate**
- **Champion, University Rover Challenge** The Mars Society, USA; Team Mongol Barota Jul. 2021 | **Certificate**
- **Second Runner-up, Anatolian Rover Challenge** UKET, Türkiye; Team Mongol Barota Jul. 2022 | **Certificate**
- **Top 10 Teams, Inter-University Hackathon** BUET CSE Fest Jul. 2023 | **Result**
- **Top 20 Teams, WIEHACK 4.0 Final Round** IEEE Women in Engineering Bangladesh Jul. 2023 | **Certificate**

Teaching and Mentoring

- **Theory Courses:** Object-Oriented Programming, Computer Architecture, Operating Systems, Structured Programming, Computer Programming, Computer Programming Techniques.
- **Laboratory and Sessional Courses:** Digital Logic Design, Structured Programming, OOP Sessional I and II, Data Structures and Algorithms Sessional II, Operating Systems Sessional, Computer Networking Sessional.
- **Supervision and Mentoring:** Co-supervise CSE-400 Undergraduate Thesis and CSE-350 Industrial Training; guided 220+ students across theory and laboratory sections through implementation, experimentation, research, and technical communication.

Leadership & Academic Service

- **Project Coordinator** May 2025 – Present
  Mobile and App Development Project for the Bangladesh Army, MIST; coordinate software development, UI/UX design, testing, and project implementation activities.
- **Moderator, MIST Cyber Security Club** Jul. 2024 – Present
  Coordinate cybersecurity workshops, CTF activities, student events, and technical operations.
- **Executive Director, MIST Computer Club** Feb. 2023 – Mar. 2024
  Organized programming competitions, technical seminars, and student-led computing initiatives.
- **Software and Communication Team, MIST Mars Rover Society** Mar. 2020 – Oct. 2023
  Contributed to rover software, communication systems, autonomous-control activities, and international rover-competition projects.

References

Dr. Muhammad Nazrul Islam
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